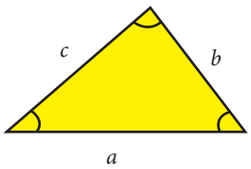
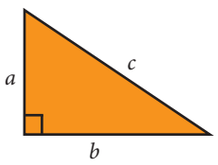
In any triangle drawn by joining three non-collinear points, the sum of the lengths of any two sides is greater than the length of the third side. This property is called as Triangle inequality.

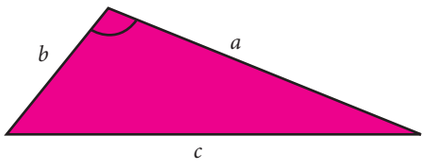
To verify this property, let us consider three types of triangles based on angles.



Acute angled triangle



Right angled triangle



Obtuse angled triangle

Fig. 4.3

For each of these triangles the following statements are true.

1. $a+b>c$ 2. $b+c>a$ 3. $c+a>b$

This property is also true for three types of triangles based on sides.

Try these

Anwser the following questions.

1. Triangle is formed by joining three _____ points.
2. A triangle has _____ vertices and _____ sides.
3. A point where two sides of a triangle meet is known as _____ of a triangle.
4. Each angle of an equilateral triangle is of _____ measure.
5. A triangle has angle measurements of 29°, 65° and 86°. Then it is _____ triangle.
   (i) an acute angled (ii) a right angled (iii) an obtuse angled (iv) a scalene
6. A triangle has angle measurements of 30°, 30° and 120°. Then it is _____ triangle.
   (i) an acute angled (ii) scalene (iii) obtuse angled (iv) right angled
7. Which of the following can be the sides of a triangle?
   (i) 5,9,14 (ii) 7,7,15 (iii) 1, 2, 4 (iv) 3, 6, 8
8. Ezhil wants to fence his triangular garden. If two of the sides measure 8 feet and 14 feet then the length of the third side is _____
   (i) 11 ft (ii) 6 ft (iii) 5 ft (iv) 22 ft
9. Can we have more than one right angle in a triangle?
10. How many obtuse angles are possible in a triangle?
11. In a right triangle, what will be the sum of other two angles?
12. Is it possible to form an isosceles right angled triangle? Explain.

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